Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 32

Elliptic Curve Cryptography-II

Estimated Reading Time: 15–20 minutes

Introduction

Elliptic Curve Cryptography – II focuses on the practical use of ECC in digital signatures, transaction authentication, and blockchain security. After understanding elliptic curves and key generation, it is important to learn how ECC enables users to sign messages and allows others to verify those signatures.

A digital signature provides evidence that a message or transaction was signed using the private key associated with a particular public key.

ECC-Based Digital Signatures

An ECC-based digital signature is a cryptographic mechanism used to authenticate messages and detect unauthorized modifications.

A widely used signature algorithm based on elliptic curve cryptography is the Elliptic Curve Digital Signature Algorithm (ECDSA).

ECDSA is used in several blockchain systems to authorize transactions.

Main Components of ECDSA

The major components involved in ECDSA are:

- Private key: The secret key used to create a digital signature.
- Public key: The corresponding key used to verify the signature.
- Message: The transaction or data that needs to be authenticated.
- Cryptographic hash: A fixed-length digest calculated from the message.
- Digital signature: Cryptographic values generated using the message digest, private key, and a securely generated per-signature nonce.

The private key must remain secret. If it is compromised, an attacker may be able to create unauthorized signatures.

Campus : Prestige Vihar, Scheme No. 74-C, Sector D, Vijay Nagar, Indore -452010 (M.P.) India
Phone : 0731-4013333, Fax : 0731-4013329 E-mail : info@piemr.edu.in, Web. www.piemr.edu.in

How ECDSA Works

ECDSA involves two major processes.

A. Signature Generation

1. The sender prepares a transaction or message.
2. A cryptographic hash of the message is calculated.
3. A secure, unique per-signature nonce is generated.
4. The signing process uses the message digest, private key, and nonce to generate the signature.
5. The signature is attached to or associated with the message.

B. Signature Verification

1. The receiver obtains the message, signature, and public key.
2. The receiver calculates the hash of the received message.
3. The verification algorithm checks whether the signature is valid for that message and public key.
4. If verification succeeds, the signature is accepted as mathematically valid.

A valid signature does not by itself establish that the transaction is legally authorized or that the signer's real-world identity is known. Those conclusions depend on the surrounding system.

Role of ECC in Blockchain Transactions

Consider a user transferring cryptocurrency to another user.

1. The sender creates a transaction.
2. The sender signs the transaction using the private key.
3. The transaction and signature are broadcast to the blockchain network.
4. Network participants verify the signature using the corresponding public key or an address-related verification mechanism.
5. If the signature and other transaction rules are valid, the transaction may be accepted for inclusion in a block.

This process helps prevent unauthorized transactions from being accepted as if they were signed by the legitimate key holder.

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ECC and Transaction Security

ECC contributes to blockchain security in several ways:

- Authentication: Helps verify that a transaction was signed using the corresponding private key.
- Integrity: Changes to signed transaction data can cause signature verification to fail.
- Authorization: Supports proving control of the cryptographic key required to authorize a transaction.
- Non-repudiation support: A valid signature can provide evidence of signing, although its legal significance depends on identity management and other circumstances.

ECC does not encrypt every blockchain transaction. Digital signatures provide authentication and integrity, whereas encryption is a separate technique used when confidentiality is required.

Security Considerations

The security of ECC depends on correct implementation and key management.

Important precautions include:

- Keep private keys confidential.
- Use cryptographically secure random-number generation.
- Never reuse ECDSA nonces across signatures.
- Protect private keys from theft and unauthorized access.
- Use well-established cryptographic libraries and appropriately selected curve parameters.

In particular, nonce reuse or predictable nonce generation in ECDSA can expose the private key.

Summary

ECC-based digital signatures help secure blockchain transactions by enabling users to sign messages with private keys and allowing network participants to verify signatures using public keys. ECDSA is a prominent example. Correct implementation, secure nonce generation, and private-key protection are essential for maintaining blockchain security.

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